

Development Automation **& Linux Lab**

Experiment 3

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EXPERIMENT-3

Shell Scripting

1. Write a shell script to fetch two command line arguments (Integers) and then add them and display the result.

Code:

```
#!/bin/bash
echo "Sum= $((($1+$2))"
```

Output:

```
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab/scripts/exp3$ ./ques1.sh 10 5
Sum= 15
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab/scripts/exp3$
```

2. Write a shell script to implement a menu-driven calculator with basic functionalities of addition, subtraction, division, and multiplication. Your script will read two numbers from users and an operation and display the result.

Code:

```
#!/bin/bash

echo "Enter:"
echo "1: Addition"
echo "2: Subtraction"
echo "3: Multiply"
echo "4: Division"

read choice

echo "Enter Two Numbers:"
read num1
read num2

case $choice in
    1)      echo "Sum= $((num1+num2))";;
    2)      echo "Diff= $((num1-num2))";;
    3)      echo "Product= $((num1*num2))";;
    4)      echo "Division= $((num1/num2))";;
    *)      echo "Wrong Choice";;
esac
```

Output:

```
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab/scripts/exp3$ ./ques2.sh
Enter:
1: Addition
2: Subtraction
3: Multiply
4: Division
3
Enter Two Numbers:
8
2
Product= 16
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab/scripts/exp3$
```

3. Write a shell script that prints all the ordinary files in the current working directory.

Code:

```
#!/bin/bash
a=$(ls)
for i in $a
do
    if [ -f $i ]
    then
        echo $i
    fi
done
```

Output:

```
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab/scripts/exp3$ ./ques3.sh
1
a.txt
abc.txt
ques1.sh
ques2.sh
ques3.sh
ques4.sh
ques5.sh
ques6.sh
ques7.sh
ques8.sh
ques9.sh
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab/scripts/exp3$
```

4. Write a shell script that takes a command line argument (the argument entered by the user is the absolute path of any file, directory, etc.,) and reports on whether it is a directory, a file, or something else.

Code:

```
#!/bin/bash
path=$1
if [ -d "$path" ]; then
    echo "Directory"
elif [ -f "$path" ]; then
    echo "File"
else
    echo "Something Else"
fi
```

Output:

```
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab/scripts/exp3$ ./ques4.sh /mnt/c/college/upes/'sem 4'/linux/lab/scripts/exp3
Directory
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab/scripts/exp3$ ./ques4.sh /mnt/c/college/upes/'sem 4'/linux/lab/scripts/exp3/abc.txt
File
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab/scripts/exp3$
```

5. Write a shell script to compute the factorial of a number entered by the user.

Code:

```
#!/bin/bash
echo "Enter Number:"
read num
fact=1
while [ $num -gt 0 ]
do
    fact=$((fact*num))
    num=$((num-1))
done
echo "Factorial: $fact"
```

Output:

```
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab/scripts/exp3$ ./ques5.sh
Enter Number:
6
Factorial: 720
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab/scripts/exp3$
```

6. Write a shell script to generate the Fibonacci series for a positive integer N which is entered by the user.

Code:

```
#!/bin/bash

echo "Enter Limit:"
read num
a=0
b=1

echo -n "$a, $b"

while [ $num -gt 2 ]
do
    c=$((a+b))
    echo -n ", $c"
    a=$b
    b=$c
    num=$((num-1))
done
echo
```

Output:

```
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab/scripts/exp3$ ./ques6.sh
Enter Limit:
10
0, 1, 1, 2, 3, 5, 8, 13, 21, 34
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab/scripts/exp3$
```

7. Write a shell script to print the value of nC_k (read as n choose k), where n and k are entered by the user in the terminal.

Code:

```
#!/bin/bash

echo "Enter value of n:"
read n
echo "Enter value of k:"
read k

facto() {
    fact=1
    for (( i=1 ; i<= $1 ; i++ ));
    do
        fact=$((fact*i))
    done
    echo "$fact"
}

echo "$(( ${facto} n / ( ${facto} $((n-k)) * ${facto} k )))" #Formula for nCr
```

Output:

```
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab/scripts/exp3$ ./ques7.sh
Enter value of n:
10
Enter value of k:
2
45
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab/scripts/exp3$
```

8. Write a shell script to find out the total instances of "the" in a given file.

Code:

```
#!/bin/bash

read -p "Enter absolute path of the file: " filepath
if ! [ -r "$filepath" ]; then
    echo "File not found!"
else
    echo "Count of 'the': $(grep -o "the" "$filepath" | wc -l)"
fi
```

Output:

```
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab/scripts/exp3$ ./ques8.sh
Enter absolute path of the file: /mnt/c/college/upes/sem 4/linux/lab/scripts/exp3/abc.txt
Count of 'the': 6
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab/scripts/exp3$
```

9. Write a shell script that displays a list of all the files in the current directory to which the user has read, write and execute permissions.

Code:

```
#!/bin/bash

for file in *
do
    if [ -r "$file" -a -w "$file" -a -x "$file" ]
    then
        echo "$file"
    fi
done
```

Output:

```
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab/scripts/exp3$ ./ques9.sh
abc.txt
ques1.sh
ques2.sh
ques3.sh
ques4.sh
ques5.sh
ques6.sh
ques7.sh
ques8.sh
ques9.sh
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab/scripts/exp3$
```